

App to Shield Communication (ATG_5)

App (PC/Mobile)

Shield (HW)

1. Initialization & Setup

INS_GET_VERSION (0x01)

[Protocol version response]

INS_INITIALIZE (0x0B)
Clear cache / initialize sensors

INS_SET_TYPE (0x07)
0 = Periodical, 1 = OnDemand

alt

[If type is Periodical]
INS_SET_TIME (0x02)
or
INS_SET_FREQUENCY (0x03)

INS_ENABLED_CHANNELS (0x04)
Active channels mask

INS_SET_SENSOR (0x0C)
Payload: hwIndex, port, sensorId...

2. Measurement (App Initiated)

INS_START (0x05)

alt

[Periodical Measurement]

loop

[Every configured interval]
Header (INS_NONE 0x00 + missed_sample) +
Channel values +
Checksum

[On-demand Measurement]

INS_GET_SAMPLE (0x08)

Header + Timestamp +
Channel values +
Checksum

3. Pause & Resume (Optional)

INS_PAUSE (0x09)

Measurement running, no data sent

INS_CONTINUE (0x0A)

4. Measurement Termination

INS_STOP (0x06)

5. Remote Control (Shield Initiated)

Shield can start/stop measurement itself (e.g., via physical button)

INS_START (0x05)

Measured data (INS_NONE)

INS_STOP (0x06)

6. Debugging

DEBUG (0x7F) +
Length + Debug
message

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Shield (HW)